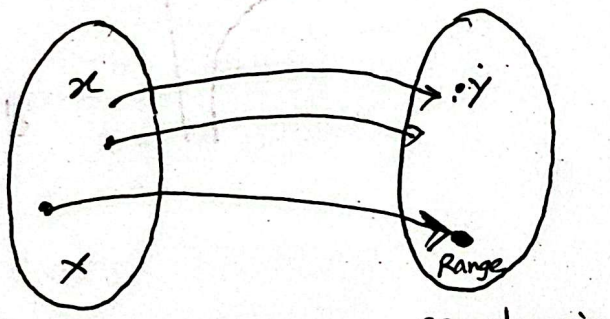
Differential calculas :

set → any relation

- function → special "
  → for each input one will get an unique output
  1 টার বেশি হবে না।

- set ও যারা engage থাকে → domain
- $\boxed{1 : many}$ → হবে না।



co-domain

We use curly brackets to substract one set from another set

ex: $\mathbb{R} - \{-2, 2\}$

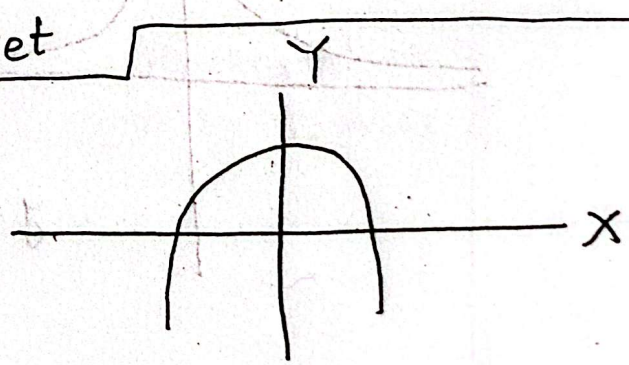
- Vertical line test: to determine if an equation is a function or not.

#

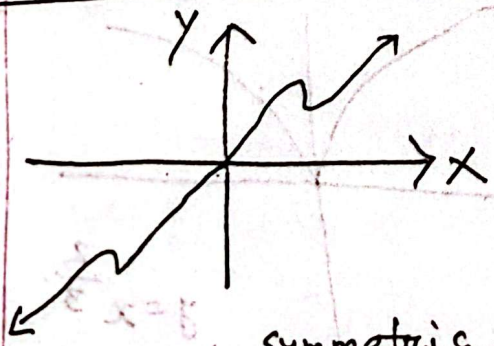
$[x \mid x \geq -4\}$ → x belongs to this set

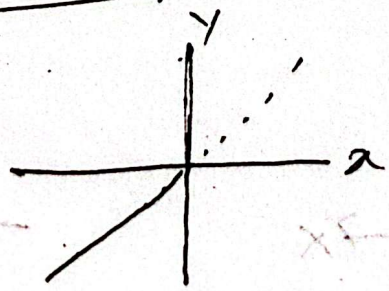
$[x \mid x \neq 5\}$

→ $[-4, 5) \cup (5, -\infty)$



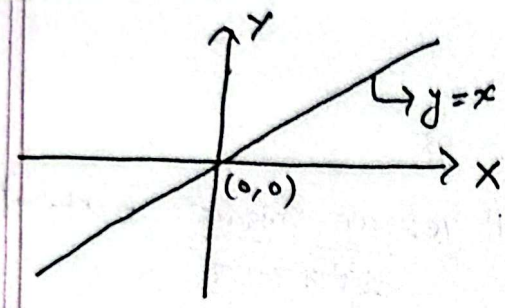
symmetric y function and it's even function

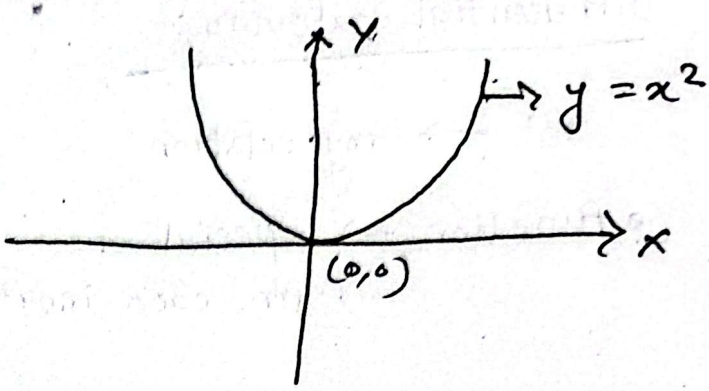


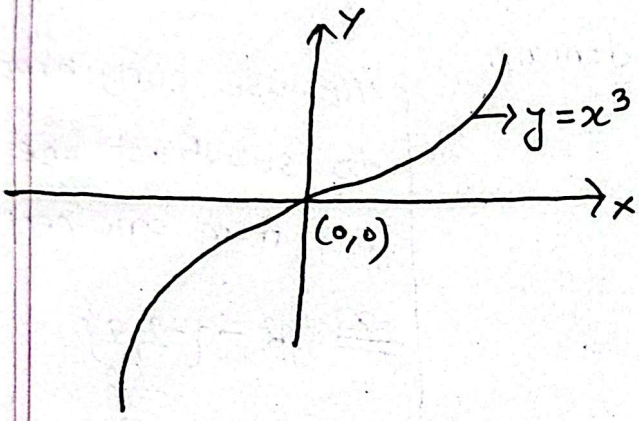


symmetric about origin and it's odd function

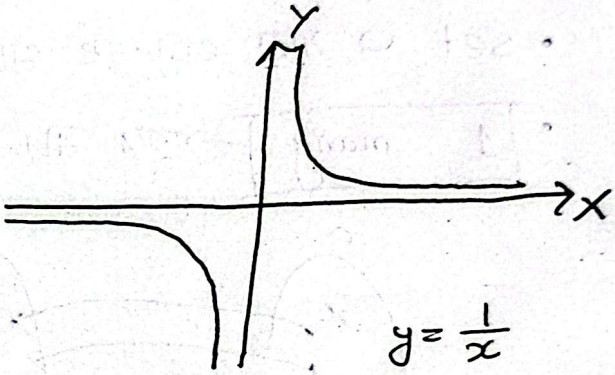
graphs:

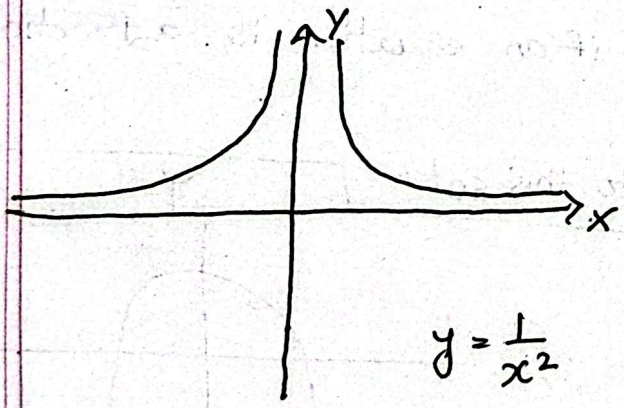


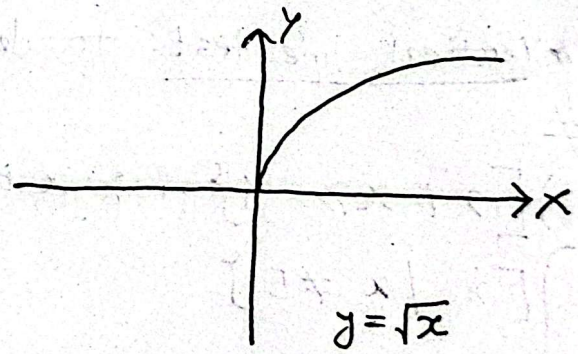


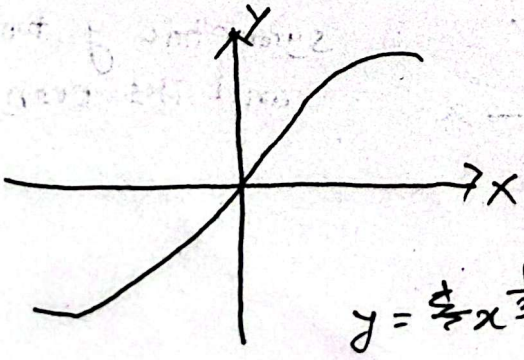


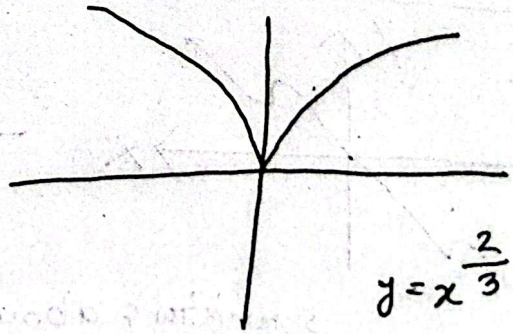
cubic curve
(s-shape)





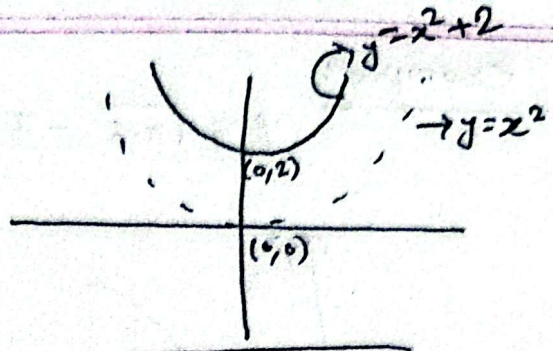




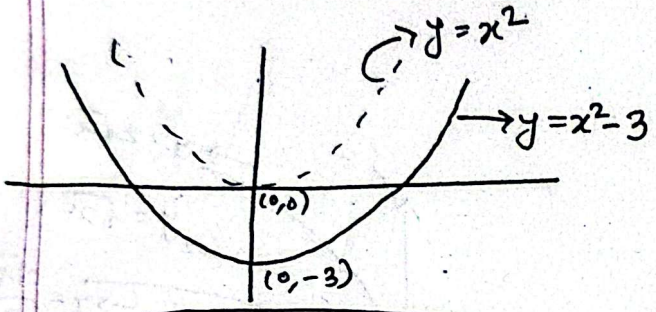


New functions from old:

$y = f(x)$	$y = x^2$
$y = f(x) + K$	$y = x^2 + 2$



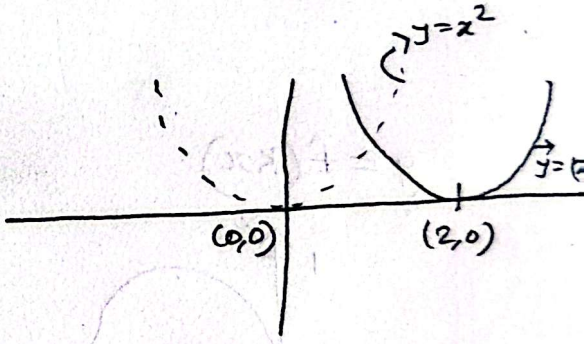
$\boxed{\text{Vertical shift}}$ up by 2 units



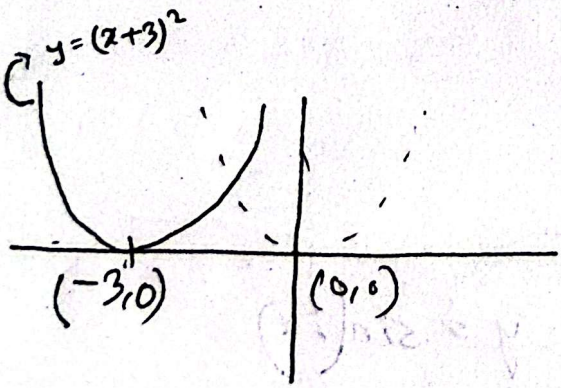
$\boxed{\text{Vertical shift}}$ down by 3 units

Horizontal Shifting:

$y = f(x)$	$y = x^2$
$y = f(x - h)$	$y = (x - 2)^2$



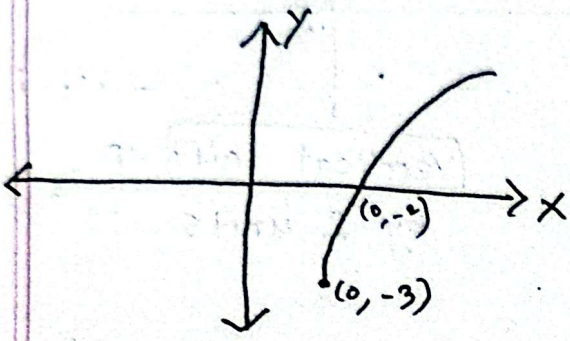
Horizontal shift right by 2 units



Horizontal shift left by 3 units

example:

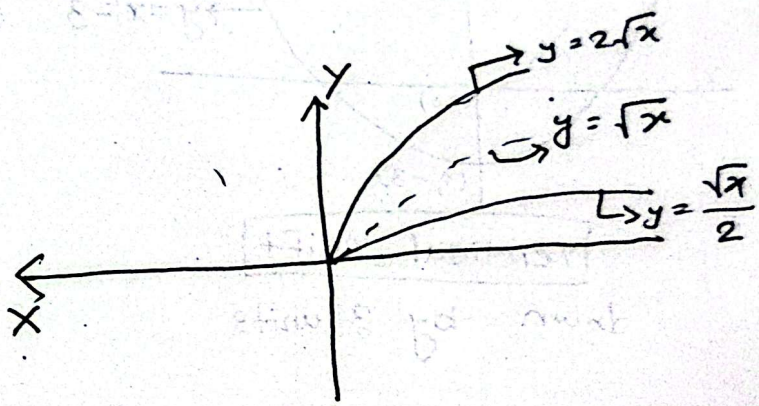
$$y = \sqrt{x-2} - 3$$



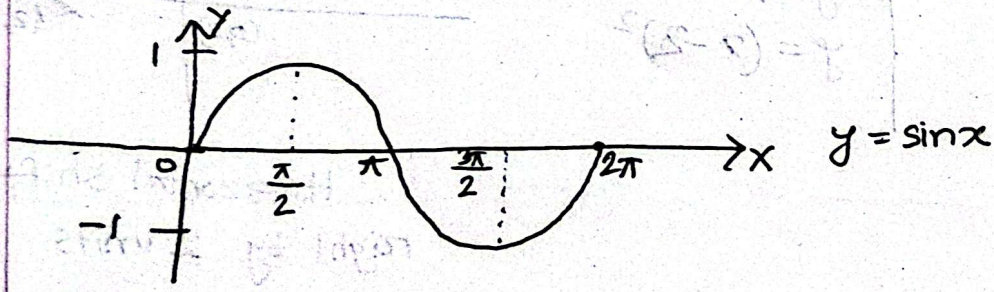
#

$y = f(x)$

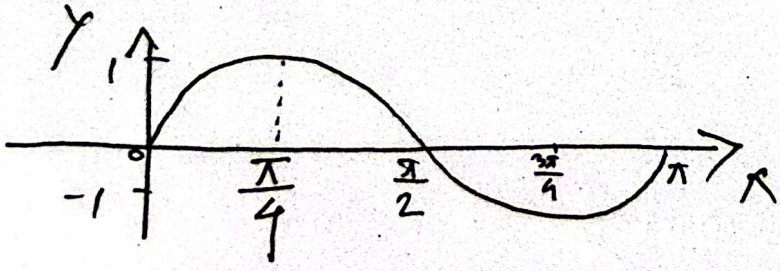
$y = k\,f(x)$ [k is positive]



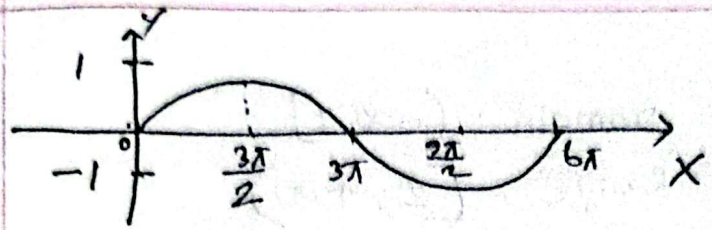
$y = f(kx)$



$y = \sin x$



$y = \sin(2x)$



$y = \sin\left(\frac{1}{3}x\right)$

Reflection about x-axis :

$y = f(x)$

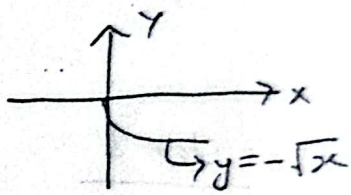
$y = -f(x)$

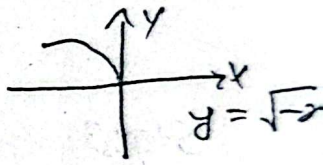
$y = f(-x)$

$y = f(x)$

$y = -f(x) \rightarrow$ reflection about x axis

$y = f(-x) \rightarrow$ " " y "

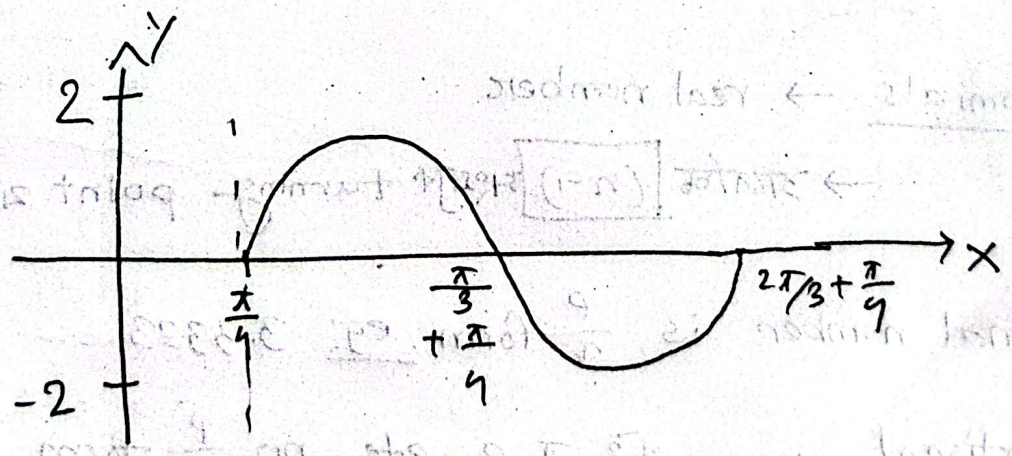




- $y = -\sqrt{-x} \rightarrow$ combination of reflection

$y = A\sin B(x - t)$

- $y = 2\sin 3\left(x - \frac{\pi}{4}\right)$



example practice:

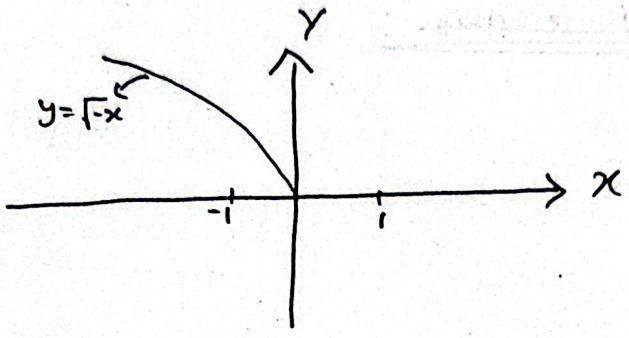
$y = f(x) = \sqrt{1-x} + 2$

$= \sqrt{-x+1} + 2$

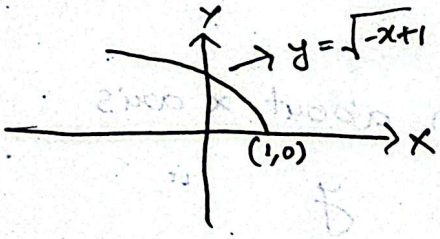
Domain : $(-\infty, 1]$

Range : $[2, +\infty)$

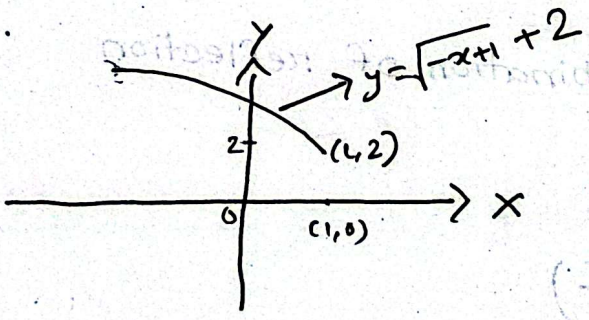
Step: 1



Step: 2



Step: 3

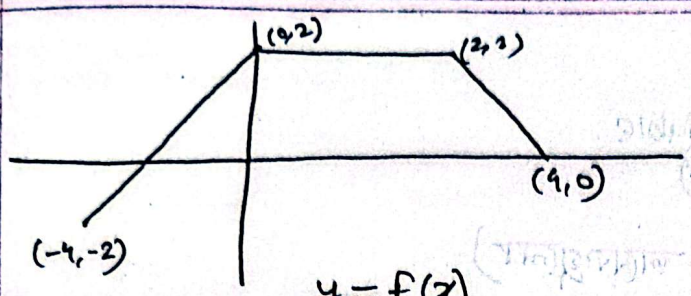


Polinomials → real number

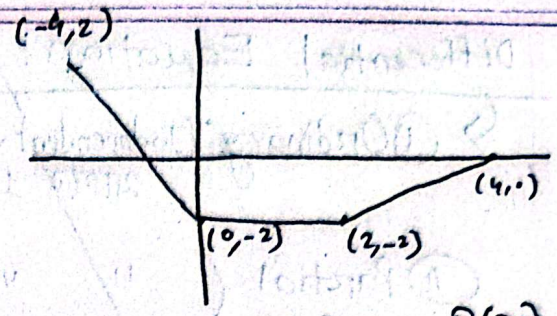
→ সর্বোচ্চ $\boxed{(n-1)}$ সংখ্যক turning point থাকবে।

Rational number is $\frac{p}{q}$ form. eg: 3.3333....

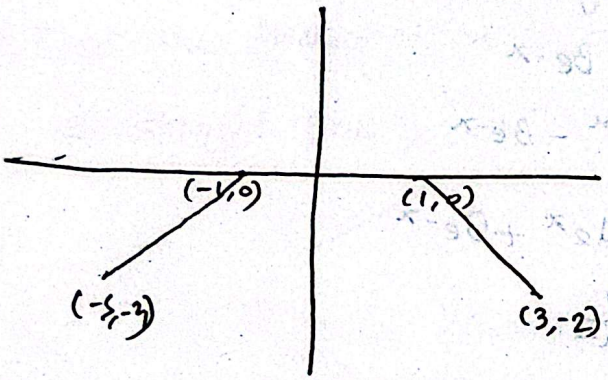
Irrational " $\sqrt{3}, \pi, e$ - etc no $\frac{p}{q}$ form.



$y = f(x)$



$p(x) = -f(x)$



$H(x) = f(x+1) - 2$

- $h(x) = f(2x)$ হয়
তাহলে $x = \frac{1}{2}$ হবে

- $g(x) = f(-x)$
↳ y axis এর সাপেক্ষে reflection হবে এবং x এর মান reflect হবে।

$y = \dfrac{k}{x^2+bx+c}$

$b = 0$

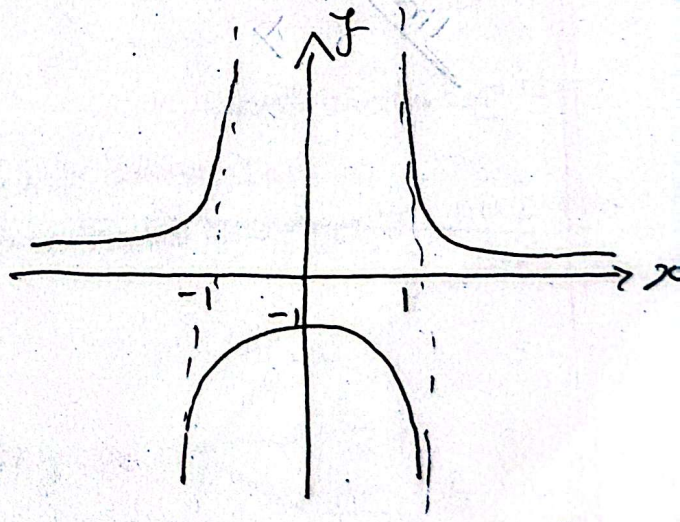
$c = -1$

$= \dfrac{k}{x^2-1}$

$-1 = \dfrac{k}{0-1}$

$k = 1$

$\therefore y = \dfrac{1}{x^2-1}$



[-1, 1]

$y = D + A\sin Bx$

$y = A\sin Bx$

Period $= \dfrac{2\pi}{B} = \dfrac{\pi}{2}$ $\therefore B = 4$ $\therefore y = 0 + (-5)\sin 4(x)$